

SELF-POWERED MUSCLE SENSOR FOR HAND GESTURE RECOGNITION

Xigan Wang

3rd Year Project Interim Report

Department of Electronic &
Electrical Engineering

UCL

Supervisor: Dai Jiang

01 December 2025

I have read and understood UCL's and the Department's statements and guidelines concerning plagiarism.

I declare that all material described in this report is my own work except where explicitly and individually indicated in the text. This includes ideas described in the text, figures and computer programs.

I acknowledge the use of ChatGPT (OpenAI, <https://chat.openai.com>) to proofread and polish the English of this report. All experimental design, circuit implementation, data analysis and conclusions are my own work.

Signed: Xigan Wang

Date: 9 December 2025

Abstract

This interim report targets hand-gesture recognition and focuses on a low-power surface electromyography (sEMG) acquisition front end and a complementary self-powered architecture. For the sEMG pathway—characterized by microvolt-level amplitudes, a limited pass-band, and a high electrode–skin source impedance—I propose a fully differential, AC-coupled instrumentation-amplifier front end with driven-right-leg (DRL) common-mode suppression and cascaded notch/low-pass filtering for robust signal extraction and power efficiency. A PCB of the analogue front end has been completed, and circuit simulations have been used to verify gain, bandwidth, and mains-frequency rejection, consistent with design targets.

On the energy side, I establish a solar-assisted self-powering framework that spans ambient light harvesting, intermediate storage, and high-efficiency power management to support continuous sEMG acquisition. To reduce battery dependence, we further explore capacitive intra-body power transfer (C-IBPT) as an energy-delivery channel, and have implemented and functionally verified both the transmitter and receiver prototypes. Together, these developments lay circuit- and system-level groundwork for a sustainable, self-powered sEMG platform.

Contents

1	Project description	2
2	Literature review	3
2.1	sEMG-based hand-gesture recognition	3
2.2	Self-powering strategy	3
3	Work performed to date	4
3.1	sEMG acquisition circuit	4
3.1.1	Biopotential Acquisition Front End	4
3.1.2	Driven Right Leg Circuit	6
3.1.3	Notch Filter(optional)	7
3.1.4	Low pass filter and AC-coupled amplifier stage	8
3.1.5	Overall Design	9
3.2	low-power design considerations, PCB design and Experimental Result . . .	10
3.2.1	Low Power Design	10
3.2.2	PCB schematic and layout	10
3.2.3	Experimental Results	12
3.3	Self-Powered Architecture	14
3.3.1	Self-Powered Architecture	14
3.3.2	Optional: Intra-body Power Transfer	15
4	Project management and lessons learned	17
5	Future work	19
	References	20
	Appendix	22
5.1	Additional simulation and measurement results	22

Section 1 Project description

The convergence of wearable electronics and the Internet of Healthcare Things (IoHT) is catalysing significant progress across domains such as motor rehabilitation, prosthetic control, and human-machine interfaces, with surface electromyography (sEMG) providing a non-invasive route to capturing forearm and hand neuromuscular activity in myoelectric control systems.[1, 2] Within this context, hand-gesture recognition has become one of the key applications of myoelectric human-machine interfaces. Existing approaches based on mechanical sensing via data gloves or vision-based methods have been widely explored; however, they depend on external hardware and line-of-sight conditions, and they show limited suitability for subtle gestures or for users with limb deficiencies or reduced muscle strength.[3, 4] Nevertheless, current sEMG wearables remain constrained by the power consumption of analogue front ends and wireless modules, and by their reliance on small lithium batteries and frequent recharging. Integrating energy harvesting, high-efficiency power management, and miniaturised storage into a self-powered wearable architecture offers a way to reduce dependence on external charging and steer myoelectric interfaces towards truly energy-autonomous operation.

In response to these challenges, this work investigates the design and implementation of a self-powered muscle sensor for hand-gesture recognition. The project concentrates on three closely related areas: (i) the development of a low-noise, high-CMRR sEMG analogue front end together with skin-comfortable electrode structures tailored for wearable deployments; (ii) the development and evaluation in MATLAB of hand-gesture recognition algorithms and feature-classification pipelines based on recorded sEMG signals; and (iii) the implementation of a self-powered power subsystem that harvests photovoltaic energy from ambient light and combines it with miniaturised energy-storage and power-management modules, in order to demonstrate the practical viability of solar-assisted operation in typical hand-use scenarios. Through coordinated optimisation of the sensing, processing, and power stages, the project aims to demonstrate the feasibility of a self-powered sEMG-based hand-gesture recognition system for long-term wear and human-machine interaction, and to summarise system-level design considerations for IoHT-oriented, battery-free myoelectric interfaces.

Section 2 Literature review

2.1 sEMG-based hand-gesture recognition

Hand-gesture recognition is a central paradigm in rehabilitation medicine, prosthetic control and assistive robotics, and has been explored using vision-based, data gloves and myoelectric sensing. Drawing on the survey by Oudah et al. [5], computer-vision-based approaches enable fully contactless interaction using RGB/IR/depth cameras, but remain sensitive to background clutter, occlusion and camera–hand geometry, which compromises robustness in unconstrained settings. Data gloves using bend, pressure and IMU sensors can capture hand posture and motion with high accuracy without relying on camera visibility [6], yet both vision- and glove-based systems are poorly suited to micro-gestures and are often inaccessible for users with limb deficiencies or restricted dexterity. In this context, surface electromyography (sEMG) provides an attractive alternative: it infers intent from neuromuscular activation rather than external motion, and therefore remains accessible to users with limited movement or amputation, making it a strong candidate for robust assistive control.

Recent research on sEMG-based gesture recognition has focused on robustness, generalization, and power efficiency. Because sEMG relies on skin-mounted electrodes, higher channel counts can improve recognition accuracy but degrade wearability and ease of use, while inevitable electrode shifts and changes in skin-conditions in daily life distort signals and undermine performance. Recent work has shown that robustness to electrode shift can be enhanced using attention deep CNNs, and that selecting a compact subset of electrodes can preserve online classification accuracy under changing wrist positions [7, 8]. Meta has further introduced a non-invasive neural motor interface based on a high-density wrist-worn sEMG band and large neural networks, enabling high-bandwidth interaction, such as in-air handwriting, without user-specific training [9].

2.2 Self-powering strategy

Wearable power strategies include piezoelectric, triboelectric and biofuel generators, as well as photovoltaic cells. However, even under idealised laboratory conditions, the reported power densities of mechanical and biochemical harvesters are typically in the range of $\mu\text{W} \cdot \text{cm}^{-2}$ to $\sim 5 \text{ mW} \cdot \text{cm}^{-2}$ —insufficient for sustaining continuous sEMG-based gesture recognition [10, 11]. By contrast, the low cost, ubiquity and relatively high efficiency of photovoltaic devices make solar cells a primary candidate for powering wearable systems, provided that harvested energy is buffered in miniaturised storage and managed by high-efficiency power electronics to support near-continuous operation. In wearables, PV placement sets incident angle, shading and bending—and thus the harvestable power. [12] To transfer energy from optimally placed solar cells to the sEMG front end on the forearm, capacitive intra-body power transfer is a promising approach. It delivers energy by injecting a high-frequency electric field across the skin, allowing power to propagate to the receiver through capacitive coupling. [13] In this project, a key trade-off arises between electrode-channel accuracy and power consumption. To balance both, I propose a three-channel, self-powered sEMG front end for reliable hand-gesture recognition.

Section 3 Work performed to date

The project encompasses the design of a low power sEMG acquisition circuit, the schematic and layout of a custom PCB, and the overall architecture of a self-powered system. This section is organised into three parts: the design of a sEMG acquisition circuit and its simulation results, low-power design considerations and the PCB design, and the architecture of the self-powered system.

3.1 sEMG acquisition circuit

The raw surface EMG (sEMG) signal is a stochastic, approximately Gaussian biopotential with typical amplitudes of about 0–10 mV_{pp} (i.e., $\approx 0\text{--}1.5\text{ mV}_{\text{rms}}$). Its usable energy lies mainly between 0 and 500 Hz, with a dominant band around 50–150 Hz, while components below ≈ 20 Hz are largely due to motion artefacts and intrinsic low-frequency instability and are therefore treated as noise. Additional noise arises from the recording electronics (broadband from 0 Hz to several kHz) and ambient electromagnetic interference, particularly 50 Hz power-line radiation whose amplitude can exceed the EMG signal by one to three orders of magnitude [14].

Nevertheless, the high source impedance at the electrode–skin interface imposes additional requirements on the sEMG acquisition circuitry. A representative architecture of the sEMG front end is illustrated in Fig. 1. The entire circuit is powered from a single 3.3 V rail and biased at mid-supply (1.65 V), established by a resistive divider network.

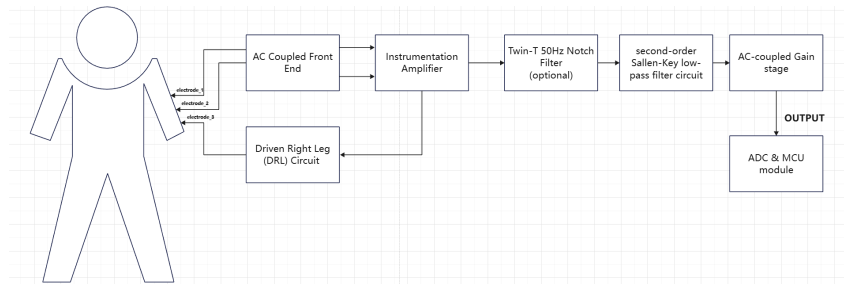


Figure 1: Block diagram of the proposed sEMG acquisition front end

3.1.1 Biopotential Acquisition Front End

For surface electromyography (sEMG), the 0-20Hz band is dominated by slow variations at the electrode-skin interface, lead motion and gross muscle movement, which manifest as low-frequency drift and motion artefacts.[14] Consequently, from a signal-characterisation perspective, the DC and ultra-low-frequency components of sEMG can be regarded as the noise. Meanwhile, the half-cell potential at the electrode-skin interface and the associated bias voltage can reach several hundred millivolts, far exceeding the average amplitude of the useful myoelectric signal.[15] If a DC-coupled input stage with high gain is used, these large DC offsets are readily amplified into saturation, preventing the sufficient gain from being realised at the front end and forcing additional gain stages downstream, thereby increasing power consumption and degrading the overall equivalent input noise.

In view of these limitations, this work adopts the AC-coupled front-end topology originally proposed by *Spinelli et al.*[15]. As shown in Figure 3, the circuit employs a fully differential passive coupling network without grounded resistors, enabling biopotential signals to be AC-coupled while keeping common-mode voltages effectively DC-coupled for regulation via the driven-right-leg (DRL) loop. By attenuating the large electrode offset potentials before they reach the high-gain stage, the architecture permits a first-stage gain of approximately 100 to be realised using only a few operational amplifiers, thereby lowering power consumption and improving input-referred noise performance.

The passive differential AC-coupling network preserves a first-order high-pass behaviour under matched components, with transfer function

$$G_{DD}(s) = \frac{s\tau_2}{1 + s\tau_2}, \quad \tau_2 = RC.$$

The corresponding cutoff frequency is

$$f_c = \frac{1}{2\pi\tau_2} = \frac{1}{2\pi RC}.$$

Here, R and C denote the resistor and capacitor of the AC-coupling network shown in Figure 3. In practice, a large R is adopted to maintain a high input impedance at the electrodes and, for a specified f_c , this choice sets a smaller C . In the implemented circuit in Figure 2, the AC-coupling resistor and capacitor are $R = 2.5 \text{ M}\Omega$ and $C = 5.6 \text{ nF}$, giving a cut-off frequency of

$$f_c \approx 11.4 \text{ Hz}.$$

Before the active processing stage, the front end requires a high-input-impedance, low-noise amplifier with stable CMRR; therefore, the INA333 is adopted as the first-stage instrumentation amplifier to accommodate the amplification of microvolt-level biopotential signals. The INA333 was selected for the AC-coupled, floating sEMG front end because it combines low input-referred noise of $50 \text{ nV}/\sqrt{\text{Hz}}$ at $G = 100$, a high common-mode rejection ratio of 100–115 dB (maintained at 50 Hz), an ultra-low input bias current (typ. 70 pA), and an extremely high differential input impedance of $\sim 100 \text{ G}\Omega \parallel 3 \text{ pF}$, together satisfying the fidelity and headroom requirements for microvolt-level biopotential amplification. [16]

Building on the AC-coupled, floating input network, the front end adopts a two-stage amplification scheme. The first stage is an INA333 whose gain is

$$G_1 = 1 + \frac{100 \text{ k}\Omega}{R_G}.$$

With $R_G = 600 \text{ }\Omega$, this gives

$$G_1 \approx 168.$$

In the test bench, assuming a differential input of 1 mV_{pp} , the first-stage output is therefore

$$V_{o1,\text{pp}} \approx G_1 \times 1 \text{ mV}_{\text{pp}} \approx 0.168 \text{ V}_{\text{pp}},$$

centred at a mid-supply bias of 1.65 V under a 3.3 V single supply.

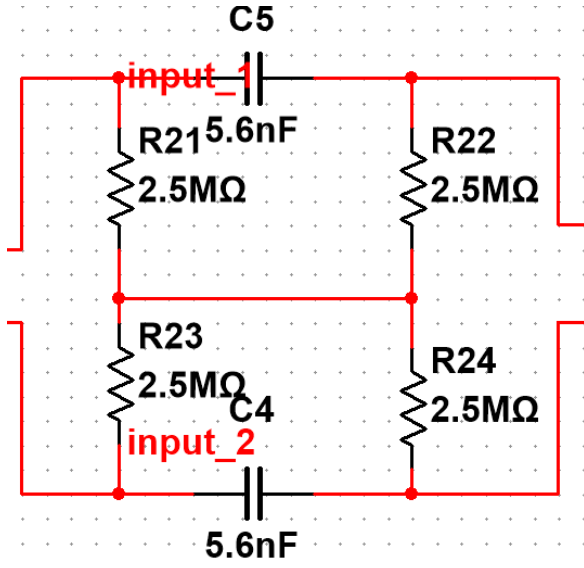
The second stage is a non-inverting amplifier whose feedback ratio is trimmed with a potentiometer to set the overall level close to, but not clipping at, the available headroom. For a target output swing of $V_{o,\text{pp}} \approx 3.1\text{--}3.2 \text{ V}_{\text{pp}}$, the required second-stage gain is

$$G_2 \approx \frac{3.1\text{--}3.2}{0.168} \approx 19\text{--}20,$$

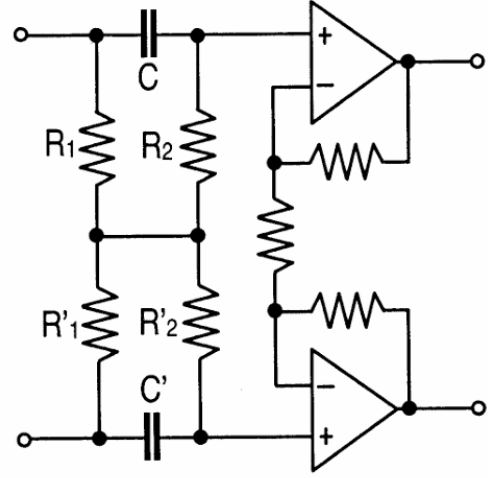
yielding a total gain of

$$G_{\text{tot}} = G_1 G_2 \approx 168 \times (19\text{--}20) \approx 3.2 \times 10^3.$$

This “fixed first stage + trimmable second stage” arrangement preserves dynamic range, avoids rail saturation, and accommodates component tolerances and test-bench variability, providing a robust amplification path for microvolt-level biopotential signals.



(a) AC-coupled differential input network (detailed view)



(b) Proposed AC-coupling circuit [15]

Figure 2: AC-coupled differential input network and proposed AC-coupling circuit.

3.1.2 Driven Right Leg Circuit

In this EMG front end, the driven-right-leg (DRL) loop is realised as an active common-mode feedback network: it senses the average voltage of the two EMG sensing electrodes, inverts and filters it, and drives the body potential via the reference electrode towards the instrumentation-amplifier. By actively reducing the common-mode voltage at the electrode–skin interface, the DRL loop effectively increases the front-end CMRR, thereby suppressing mains-induced 50Hz ripple on the EMG signal.

The experimental results are summarised in Fig. 3. Two 100 Hz sinusoidal inputs of 2 mV_{rms} and 1 mV_{rms} with a phase difference of 180° are applied to generate a differential test signal, while a 0.1 V_{rms}, 50 Hz voltage is injected equally on both inputs to emulate common-mode interference, whose magnitude is two orders of magnitude larger than the test signal. With the DRL circuit enabled, the output frequency spectrum in Fig. A2(a) (left) contains only the 100 Hz component, indicating that the 50 Hz common-mode interference has been effectively suppressed. In the absence of the DRL circuit, the spectrum in Fig. A2(b) (right) becomes a superposition of the 50 Hz and 100 Hz components. Taken together, the spectra in Fig. A2 demonstrate that the DRL loop provides strong attenuation of the 50 Hz mains-frequency common-mode noise.

In the implemented front end, the DRL path is realised with a two-stage amplifier based on the AD8542 (U7B, U2B). The design of this DRL circuit is proposed by *Spinelli et al.*[15]. The common-mode voltage at the INA333 inputs is first sensed by a resistive network and referred to the INA333 REF node; this error signal is then processed by U7B, which is configured as an inverting leaky integrator. The inverting input of U7B is driven through $R_{13} = 10 \text{ k}\Omega$, while the feedback network consists of $R_{12} = 510 \text{ k}\Omega$ in series with $C_6 = 10 \text{ nF}$. The small-signal transfer function of this stage is therefore

$$A_{\text{DRL}}(s) = -\frac{Z_f}{R_{13}} = -\frac{R_{12}}{R_{13}} - \frac{1}{sR_{13}C_6},$$

which combines a proportional term $-R_{12}/R_{13} \approx -51$ with an integral term $-1/(sR_{13}C_6)$. At very low frequencies the integral term dominates, so that any quasi-DC error between the body potential and the REF node is accumulated over time and actively driven towards zero. This integral

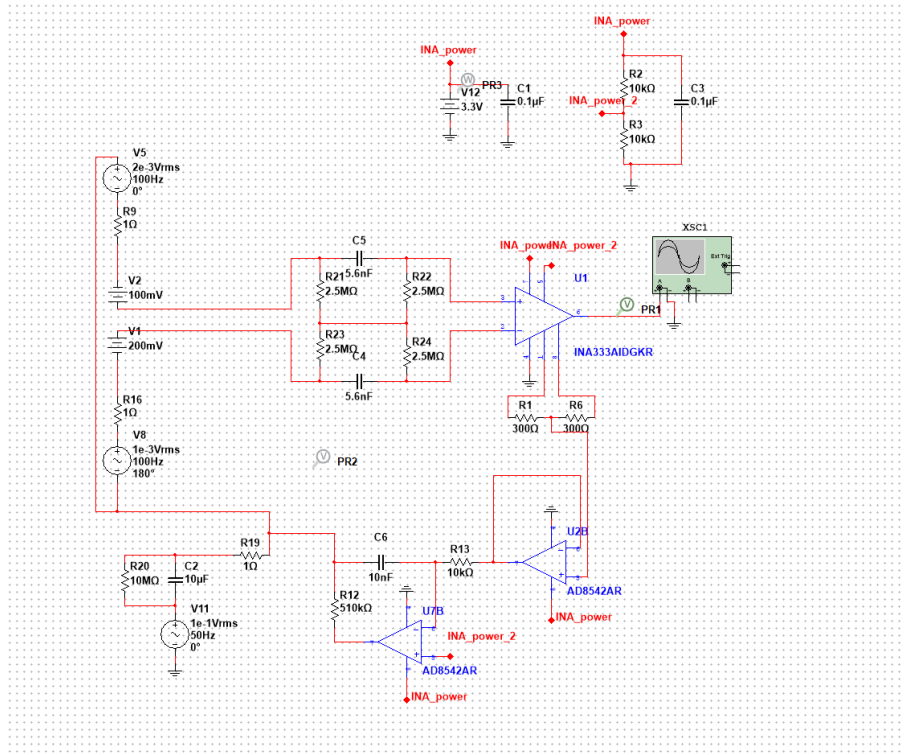


Figure 3: Driven-Right Leg Circuit Test Simulation

action strongly suppresses the half-cell potentials and slow drift at the electrode–skin interface, and provides high loop gain around the 50 Hz mains frequency. The series combination of R_{12} and C_6 introduces a pole at

$$f_p \approx \frac{1}{2\pi R_{12} C_6} \approx 31 \text{ Hz},$$

so that above a few tens of hertz the contribution of the integral term rolls off and the DRL gain asymptotically approaches the finite value of -51 . In the EMG signal band (hundreds of hertz), the DRL loop therefore exerts only modest feedback, which avoids distortion of the differential EMG component and improves phase margin in the presence of the body and lead capacitances.

The output of U7B is buffered by a unity-gain stage U2B and applied to the driven-right-leg electrode through a large series resistor to limit patient current. In this way the loop injects an inverted, band-limited version of the sensed common-mode voltage back into the body, tightly regulating the common-mode level at the INA333 inputs while maintaining stability and preserving the desired EMG bandwidth.

3.1.3 Notch Filter(optional)

Power-line interference at 50 Hz can, in principle, be mitigated using a notch filter; however, a conventional notch not only removes 50 Hz but also attenuates neighbouring frequencies, risking distortion of sEMG content within 50–150 Hz[15]. Accordingly, it is preferable to suppress the disturbance in the common-mode domain before high gain: a driven-right-leg (DRL) loop lowers the electrode–skin common-mode voltage; moreover, because mains interference typically appears at the two inputs with near-equal amplitude and in phase, the instrumentation amplifier’s high CMRR can further reject the residual 50 Hz component. If necessary, a narrowly tuned notch may be retained as a final stage, but its Q and bandwidth must be chosen carefully to avoid encroaching on the passband of interest.

An active twin-T notch is well-suited for 50 Hz suppression because its feedback-tunable Q enables a narrow, deep notch at 50 Hz while minimally encroaching on the 50–150 Hz sEMG band. The op-amp buffering isolates source/load effects and improves tolerance to component spread, complementing DRL and the INA's high CMRR for layered mains-noise rejection.

Figure 4(b) illustrates a classic notch-filter design, in which the notch frequency f_N , quality factor Q , and feedback fraction k are all tunable via the chosen resistor and capacitor values (R and C). The function of the notch centre frequency is

$$f_N = \frac{1}{4\pi RC}.$$

With $R = 16 \text{ k}\Omega$ and $C = 100 \text{ nF}$, this yields

$$f_N = \frac{1}{4\pi \times 16000 \times 100 \times 10^{-9}} \approx 49.8 \text{ Hz}.$$

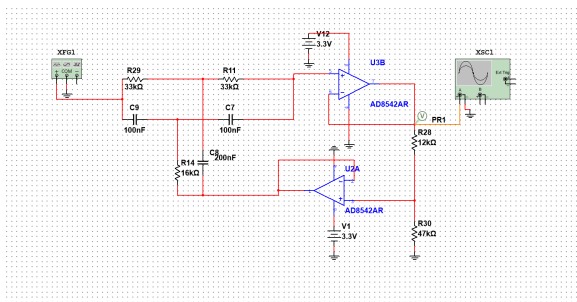
making it well-suited for suppressing 50 Hz mains-frequency interference.

As noted in the TI application report, high- Q notch realisations such as the classic twin-T are difficult to implement in practice because the resistor ratios that set Q become extremely critical at high values of Q , and op-amp stability (phase margin) is easily compromised when the effective feedback approaches unity [17].

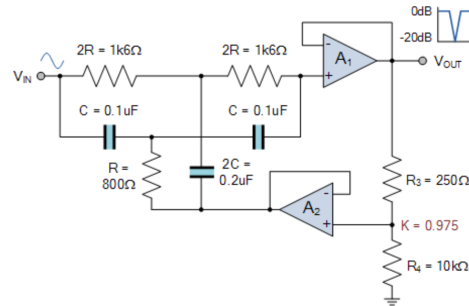
As shown in Fig. A3, the bottom case yields $Q = 0.5$, while the top case yields $Q = 4.75$. Accordingly (for $f_N = 50 \text{ Hz}$), the -3 dB bandwidths are $\text{BW} = 100 \text{ Hz}$ and $\text{BW} \approx 10.5 \text{ Hz}$, respectively. This demonstrates that bandwidth is inversely related to Q : a larger Q gives a narrower notch.

Therefore, for $Q = 4.75$, the relation $k = 1 - \frac{1}{4Q}$ gives $k = 1 - \frac{1}{19} = \frac{18}{19} \approx 0.947$. This corresponds to a divider ratio of 18:1, for which 18 k Ω and 1 k Ω satisfy the requirement.

Meanwhile, considering power consumption, stability, and collateral attenuation of adjacent spectral components, delegating the remaining mains suppression to the digital domain is often preferable: a post-ADC digital notch (or adaptive) filter can remove the 50 Hz component more efficiently and robustly while preserving the surrounding sEMG band. Whether the analogue notch is required will be decided after the testbench evaluation and first-revision measurements; if DRL and the INA's CMRR already meet the mains-rejection target, we will omit it and, if necessary, apply a post-ADC digital notch instead.



(a) Fig. 4A. 50 Hz twin-T notch filter design with a 1.65 V DC offset.



(b) Fig. 4B. Example implementation for the 50 Hz twin-T notch filter.

Figure 4: Twin-T notch filter at 50 Hz: (A) circuit design with DC offset; (B) example realisation and component choices.

3.1.4 Low pass filter and AC-coupled amplifier stage

For sEMG signals, additional low-pass filtering and AC-coupled amplification are required. Figure 5 illustrates the unity-gain Sallen-Key second-order low-pass filter used in this work, whose cut-off

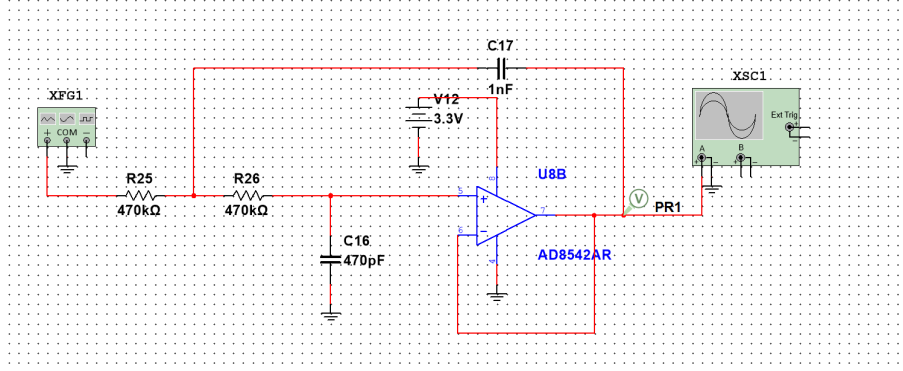


Figure 5: Sallen-Key second-order low-pass filter

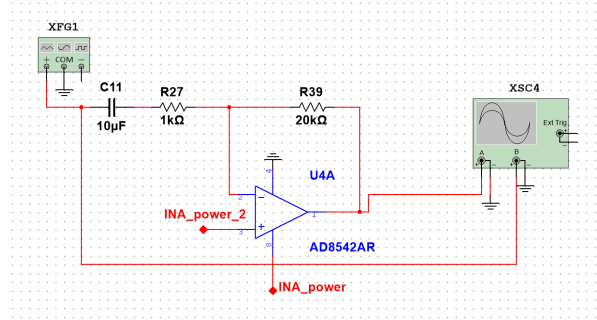


Figure 6: 2nd stage amplifier

frequency is given by

$$f_c = \frac{1}{2\pi\sqrt{R_{25}R_{26}C_{16}C_{17}}}.$$

Substituting $R_{25} = R_{26} = 470 \text{ k}\Omega$, $C_{16} = 470 \text{ pF}$, and $C_{17} = 1 \text{ nF}$ gives $f_c \approx 494 \text{ Hz}$, and the corresponding frequency spectrum is shown in Figure A4.

Voluntary sEMG power is largely confined to the 20–450 Hz band, so choosing a cut-off slightly above this range preserves the relevant myoelectric components while attenuating high-frequency amplifier noise and interference from surrounding digital electronics. The chosen capacitor ratio $C_{17}/C_{16} \approx 2.1$ results in a relatively low quality factor ($Q \approx 0.28$), giving a strongly damped second-order response without passband peaking. This avoids overshoot and ringing in the time domain, which is desirable for accurately capturing rapid changes in muscle activation.

A second-stage inverting amplifier based on the AD8542 is used to provide additional gain. The circuit is referenced to the mid-supply bias node, and its small-signal gain is set by

$$A_v = -\frac{R_f}{R_{in}}.$$

In the final hardware implementation, the fixed feedback resistor R_f is replaced by a trim potentiometer, so that the overall gain can be adjusted on the bench to accommodate variations in sEMG amplitude and to avoid output saturation during strong contractions.

3.1.5 Overall Design

Figure 7 shows the overall circuit design, and Figure A5 depicts the resulting frequency spectrum, confirming that the salient features of the sEMG signal are effectively amplified.

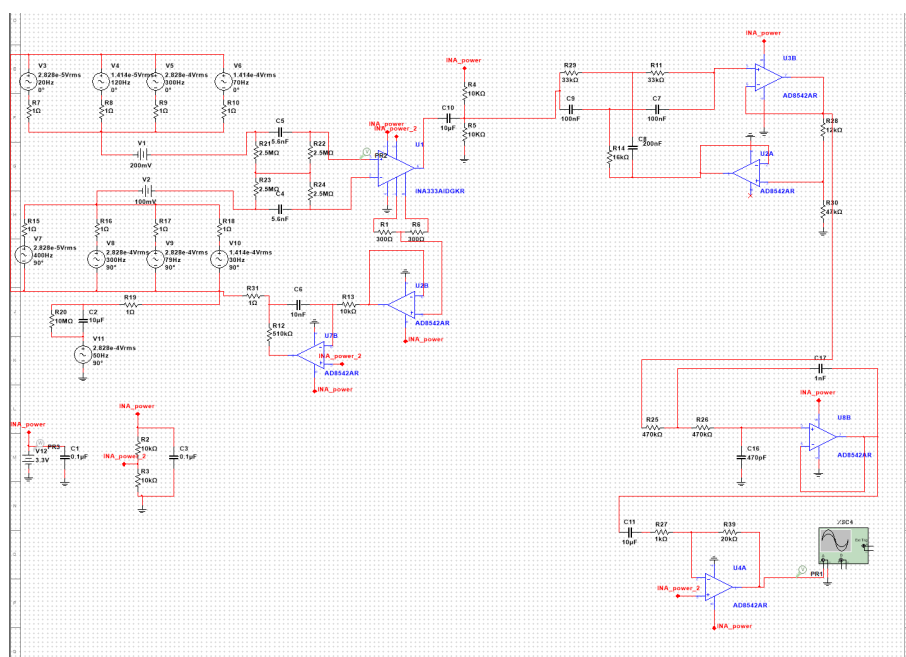


Figure 7: Simulation Design

3.2 low-power design considerations, PCB design and Experimental Result

3.2.1 Low Power Design

As the entire circuit is required to operate in a self-powered manner, low-power operation must be prioritised in the design. In simulation, the average power consumption is approximately 2.28 mW per channel; in practice, this value is expected to be higher owing to the finite conversion efficiency of the LDO and other non-ideal factors. The overall circuit is optimised at both the power-supply level and the component level by simplifying the circuitry and selecting energy-efficient components.

In this design, a 3.3 V single-supply configuration is adopted to minimise power consumption. Assuming a comparable supply current, operating the op-amp from a $\pm 5\text{ V}$ supply (i.e. a 10 V span) would dissipate roughly three times the quiescent power of the 3.3 V single-supply configuration, even before accounting for the additional losses of the negative-rail generation circuitry. The 1.65 V mid-supply reference is generated by a resistor-divider network from the 3.3 V rail.

In this design, both the INA333 instrumentation amplifier and the AD8542 dual op-amp were chosen for their very low quiescent currents: the INA333 draws approximately $50\mu\text{A}$ and the AD8542 about $45\mu\text{A}$ per amplifier, corresponding to only $\sim 0.17\text{mW}$ and $\sim 0.15\text{mW}$ per channel, respectively, at a 3.3V supply. [18, 19].

3.2.2 PCB schematic and layout

As shown in Figure 8, the complete schematic of the proposed design is presented. Figures 9 and 10 depict the PCB layout of the board, illustrating the top and bottom layers, respectively.

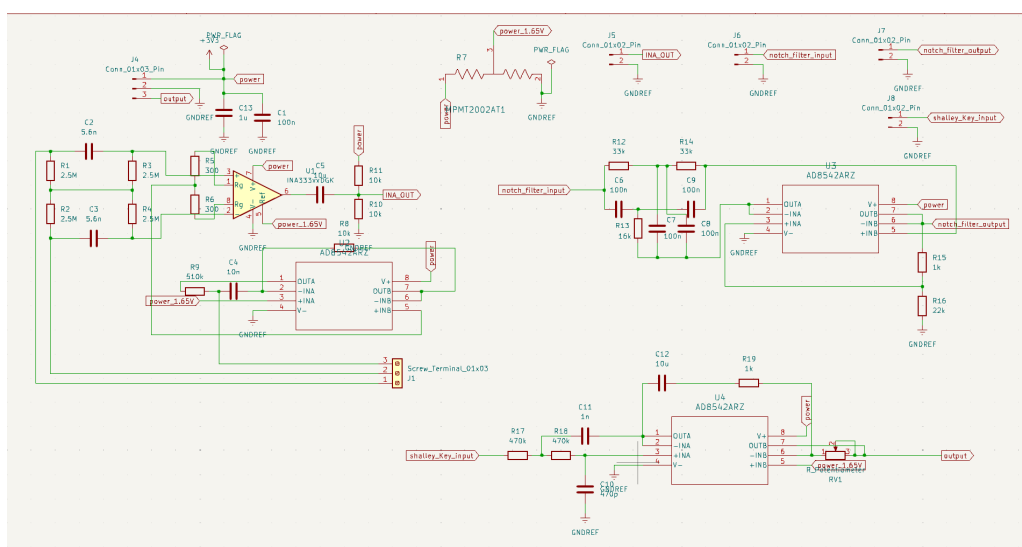


Figure 8: Complete schematic of the proposed design

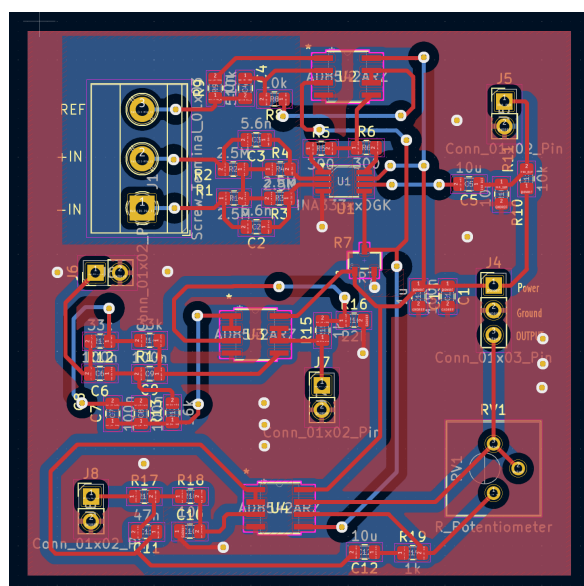


Figure 9: Top-layer PCB layout

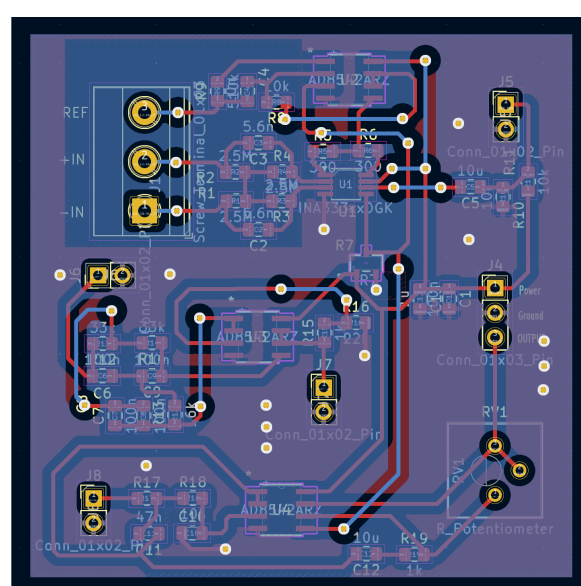
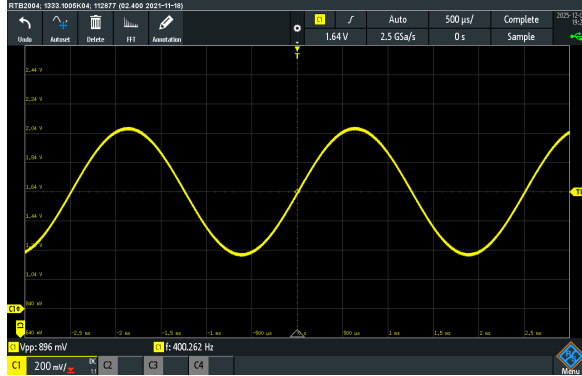


Figure 10: Bottom-layer PCB layout

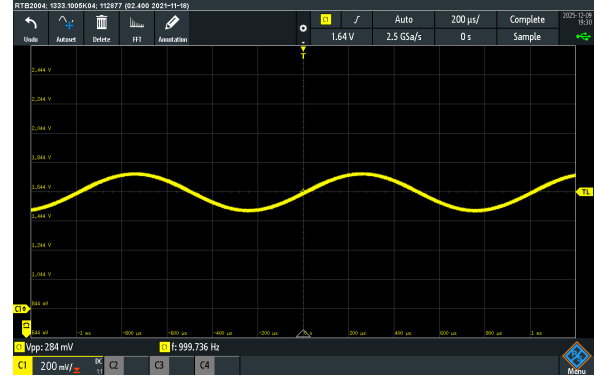
The PCB is implemented as a two-layer board and adopts a compact 6×6 cm form factor tailored for wearable use. To satisfy the mechanical and layout requirements of wearable integration, the power interface and signal input are intentionally placed on opposite sides of the PCB. The input is connected via a screw terminal and flying leads, whereas the power supply and output are interfaced through header pins. The overall circuit is implemented with a modular architecture: the intermediate twin-T notch filter is connected via header pins, allowing it to be included or bypassed in order to evaluate its impact on the overall system performance. The voltage division is provided by the voltage-divider network. In terms of PCB routing, particular care was taken around the highly sensitive sEMG input. The front-end AC-coupling network was implemented with a symmetric differential layout, and the INA333 placement and routing closely follow the datasheet recommendations in order to maximise gain accuracy and minimise noise pickup.

3.2.3 Experimental Results

In practical bench testing, the fabricated PCB was fully assembled and soldered, and evaluated using a signal-generator-based testbench. During these tests, the two input terminals were driven from the signal generator, while the DRL output was temporarily tied to ground.

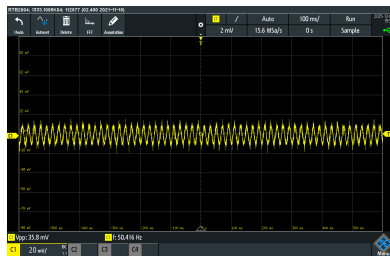


(a) Low-pass filter output at 400 Hz

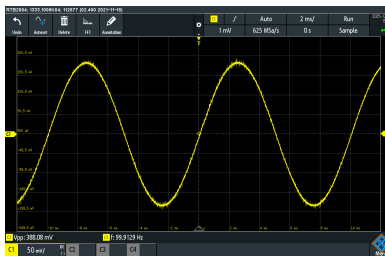


(b) Low-pass filter output at 1 kHz

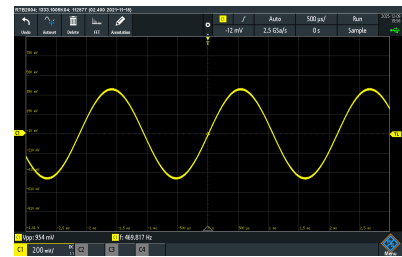
Figure 11: Low-pass filter responses at different input frequencies



(a) 50 Hz notch

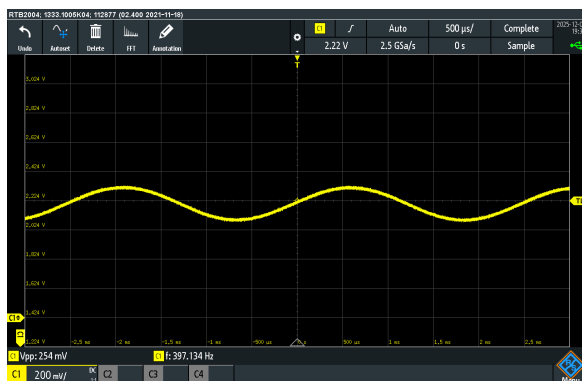


(b) 100 Hz notch

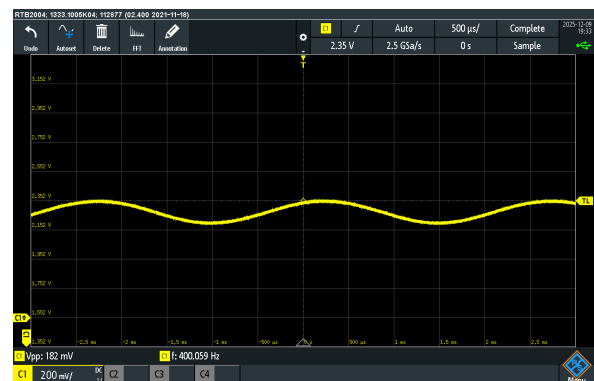


(c) 470 Hz notch

Figure 12: Notch filter responses at different centre frequencies



(a) 400 Hz with 250 Ω feedback resistor



(b) 400 Hz with reduced feedback resistor

Figure 13: AC-coupling Amplifier

Figures 11–13 show the experimental results for the low-pass filter, the passive notch filter and the AC-coupled amplifier, respectively. For the low-pass filter (Figure 11), with a 1 V_{pp} input, the output at 400Hz is close to 1 V_{pp} , whereas at 1kHz it decreases to about 280 mV_{pp} . This corresponds to

an attenuation of approximately -11.1 dB, which is within about 1.5 dB of the theoretical -12.6 dB and can be attributed to normal experimental error.

For the passive notch filter (Figure 12), we applied a 1 V_{pp} input and evaluated the response around the 50 Hz notch frequency. At the centre frequency, the output is approximately $35.8\text{ mV}_{\text{pp}}$, corresponding to a suppression of about -28.9 dB relative to the 1 V_{pp} input, indicating good notch performance at 50 Hz . However, at 100 Hz the amplitude is still reduced to around $388\text{ mV}_{\text{pp}}$, and the waveform only recovers to near its full amplitude at roughly 470 Hz . This behaviour highlights the influence of the quality factor Q on the amplitude of frequency components in the vicinity of the notch.

For the final AC-coupled amplifier stage (Figure 13), the gain is set by an inverting configuration with

$$A_v = -\frac{R_{\text{trimmer}}}{R_{\text{in}}}, \quad R_{\text{in}} = 1\text{ k}\Omega,$$

where the trimmer has an adjustable range of $0\text{--}250\text{ }\Omega$. Within the effective frequency range defined by the preceding low-pass filter, varying R_{trimmer} produces an approximately linear change in the output amplitude, confirming that the stage behaves as expected. Nonetheless, the current design could be improved by selecting a trimmer with a higher resistance or by modifying R_{in} to achieve a more suitable gain range; the observed controllable gain still demonstrates that the overall system is functioning correctly.

The main problem occurs in the first gain stage. In this stage, the output exhibits the same frequency components as the input, but the expected DC bias is not properly reflected; I will address this in subsequent experiments. In addition, the present voltage-divider design uses only a simple two-resistor network, so the divided node is pulled up by the rest of the system. In the next design iteration, the supply network will therefore be implemented using a buffer and an LDO regulator to provide a stable bias.

3.3 Self-Powered Architecture

3.3.1 Self-Powered Architecture

As outlined in the literature review, miniature solar cells are selected as the most suitable candidates for the energy-harvesting scheme, primarily due to their relatively high energy-conversion efficiency and the fact that such devices are relatively easy to obtain.

In the proposed power architecture, the solar cell continuously harvests energy from ambient light and feeds it into an energy-harvesting IC, which performs the necessary voltage conversion and charging management. The harvested energy is used in the first instance to supply the load, while any surplus is stored in a rechargeable battery or storage capacitor. When the illumination level is low or absent, the power-path management circuitry automatically draws energy from the battery side to maintain a stable supply voltage for the sEMG front-end. In this way, the solar cell acts as the primary energy source under illuminated conditions, whereas the battery serves as an energy buffer that ensures continuous operation during intermittent or low-light periods.

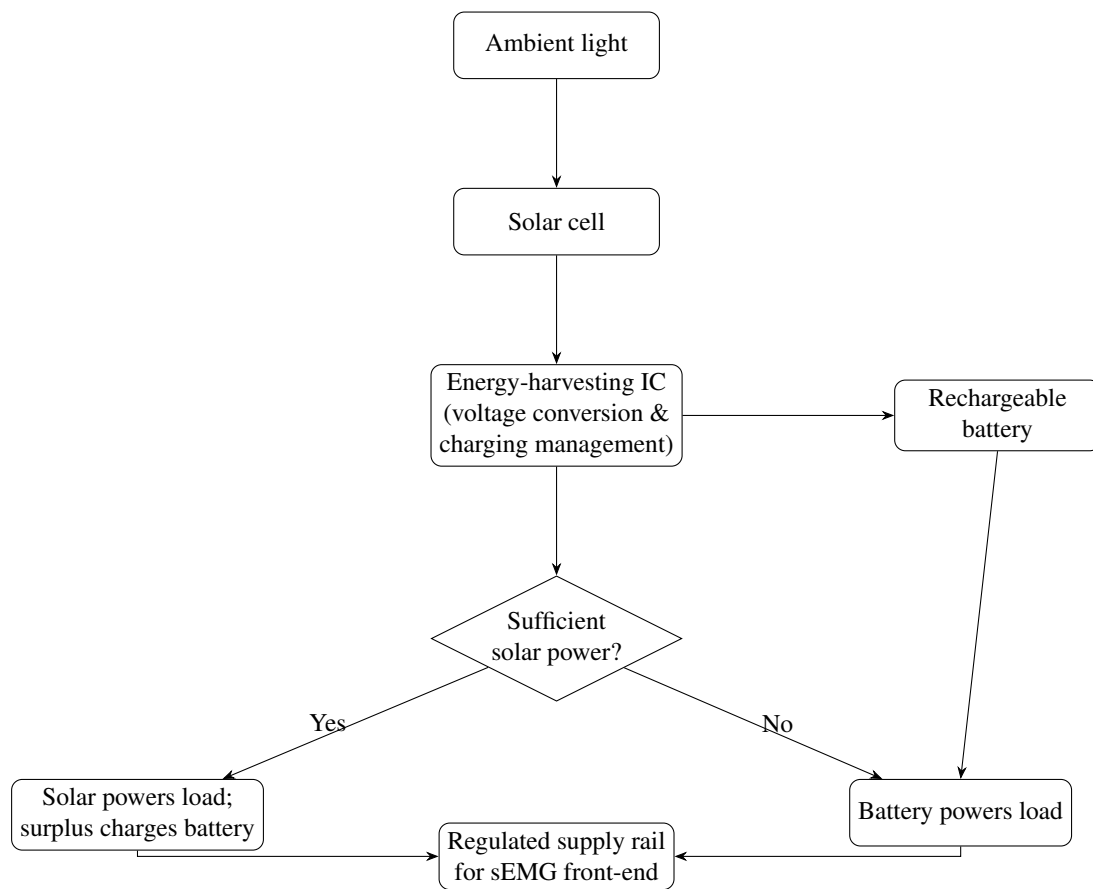


Figure 14: Overall power-flow diagram for the solar-cell and battery hybrid supply

Two representative integrated circuits for hybrid solar–battery power management are the BQ25570 and the MCP73871. The BQ25570 is an ultra-low-power energy-harvesting PMIC that integrates a boost front end with MPPT, storage management and a high-efficiency buck output, making it well suited to extracting energy from low-level, high-impedance sources such as small solar cells. In this design, the MCP73871 is treated as a representative linear Li-ion charger that implements power-path and load-sharing control between the regulated solar input and the Li-ion storage battery.[20]. [21] For wearable systems operating from low and highly variable input power, the

BQ25570 therefore offers better performance and is more appropriate for deployment in changing environmental conditions.

3.3.2 Optional: Intra-body Power Transfer

Intra-body power transmission provides an additional energy-delivery path over the skin, allowing the solar cells in the self-powered system to be placed purely according to illumination and power-extraction efficiency rather than wiring constraints. Specifically, capacitive intra-body power transfer (C-IBPT) is a body-coupled powering technique that exploits the capacitive channel between the human body and its environment to deliver energy at MHz-range carrier frequencies, analogous to capacitive HBC links but optimised for power rather than data: an external supply drives an RF signal through a power amplifier and impedance-matching network into body electrodes at the transmitter, while the receiver uses an LC resonant circuit, a Schottky-diode voltage doubler and an energy-harvesting PMIC (e.g. BQ25570) to rectify, store and regulate this RF energy into a stable DC rail for wearable or on-skin electronics. Figure 15 shows this C-IBPT system architecture.

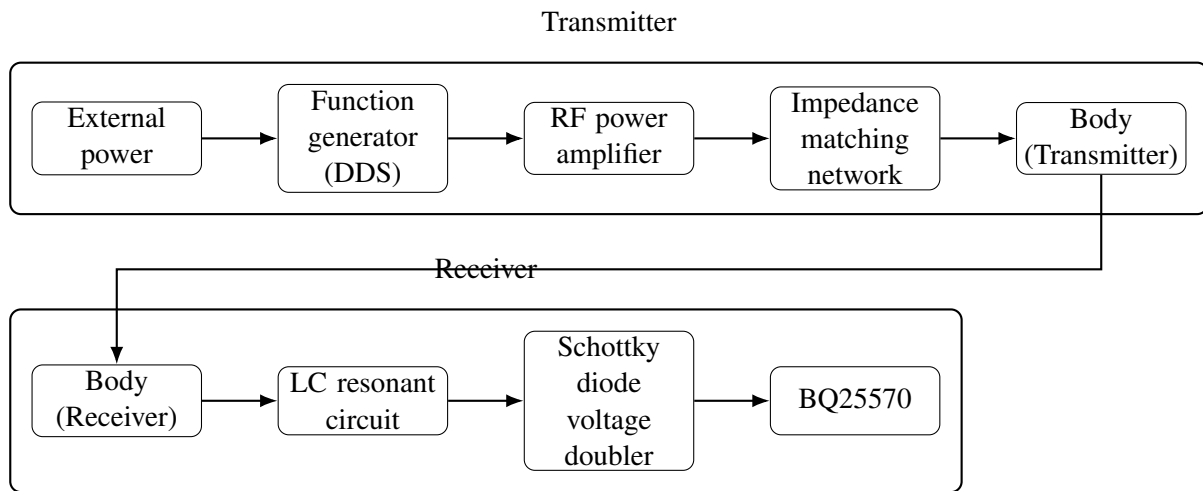
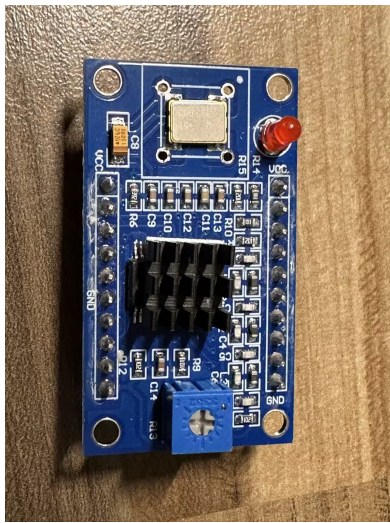


Figure 15: Block diagram of the C-IBPT architecture .

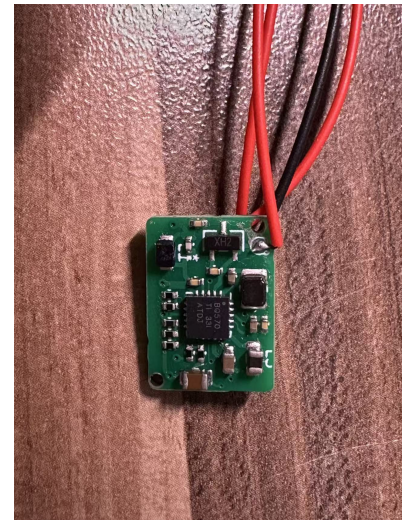
In the experimental prototype, an Arduino microcontroller configures and drives an AD9850 direct-digital-synthesis (DDS) module to generate a 40 MHz sinusoidal carrier for the C-IBPT link. Because the AD9850 can only provide a small output swing of approximately 10 mV_{pp} at tens of megahertz, the RF signal is first boosted by an external RF power amplifier and then delivered to the body through metal electrodes placed on the skin, thereby exciting the capacitive intra-body channel. In the current benchtop setup, the power arriving at the skin electrodes is still relatively low due to the limited drive capability of the modules and the lack of a dedicated impedance-matching network; future revisions could address this by introducing an optimised matching network and/or employing a higher-swing DDS or RF source. On the receiver side, a tuned LC resonant circuit is centred around the 40 MHz carrier to increase the voltage across the receive electrodes; a Schottky-diode voltage doubler then performs high-efficiency RF–DC rectification. The rectified voltage is subsequently fed into an energy-harvesting PMIC (BQ25570), which manages energy storage and regulates the output to a usable DC level. In measurements, when a $500\text{ mV}_{\text{pp}}$, 40 MHz sinusoid is applied at the receiver input, the prototype achieves an open-circuit DC output of approximately 3 V, demonstrating the feasibility of using a capacitive intra-body link with off-the-shelf DDS and RF amplifier modules to deliver power to on-skin or wearable electronics.



(a) AD9850 DDS module



(b) RF power amplifier



(c) BQ25570 energy-harvesting module

Figure 16: Prototype implementation of the C-IBPT testbench, including the DDS-based transmitter (AD9850), RF power amplifier and BQ25570-based energy-harvesting receiver.

Within the overall system, this C-IBPT link is intended as a supplementary power-delivery path rather than a replacement for the primary solar self-powered architecture and sEMG acquisition/processing circuitry: once the core self-powered front-end and signal-processing chain are in place, adding this intra-body power-transfer channel can further relax constraints on module placement and wiring, thereby improving the overall form factor and long-term wearability of the device.

Section 4 Project management and lessons learned

Progress against original plan and updated scope Compared to the original Gantt chart, the project is broadly on schedule. The initial milestones were to design the sEMG acquisition circuit, complete the PCB layout and fabrication, and bring up a basic testbench. These milestones have been followed, but one important adjustment has been made to the project scope: the C-IBPT link is now treated as an optional extension rather than a core deliverable. The current plan is to prioritise the realisation of a stable self-powered architecture for the sEMG front end, and only then investigate integrating C-IBPT as a supplementary power-delivery path to further improve wearability. A key technical risk at this stage is that the INA333 on the testbench does not currently provide the expected gain in measurements. This issue will be investigated in the next round of testing to identify whether the root cause lies in the biasing, component values, PCB layout or measurement setup; if necessary, a second PCB revision will be designed at the end of Term 1 to address any confirmed design issues.

Lessons learned The work completed so far has highlighted several lessons. First, the time required to order and receive critical components was underestimated: lead times and stock constraints caused approximately a one-week delay relative to the original schedule. Second, parts of the analogue design did not sufficiently account for real-world component tolerances (for example, resistor and capacitor value spreads in biasing and filtering networks), which can shift pole and zero locations and potentially introduce instability or degraded performance in the implemented circuit compared with the ideal simulations.

Implementation plan and updated Gantt chart To address these issues in Term 2, a more proactive planning and verification strategy will be adopted. The next iteration of the analogue front end and self-powered architecture will be completed over the vacation period, so that PCB and component orders can be placed at the very start of the term, with explicit “plan B” options (alternative parts and suppliers) identified in advance. At the simulation stage, each critical sub-circuit will be validated from multiple perspectives, including parameter sweeps and tolerance analysis, to check that the design remains robust under realistic component variations. Figure 17 presents the updated Gantt chart: the main Term 2 activities are now structured around implementing and validating the self-powered architecture, while the C-IBPT link has been moved to a later phase of the project as an optional extension block rather than a core dependency.

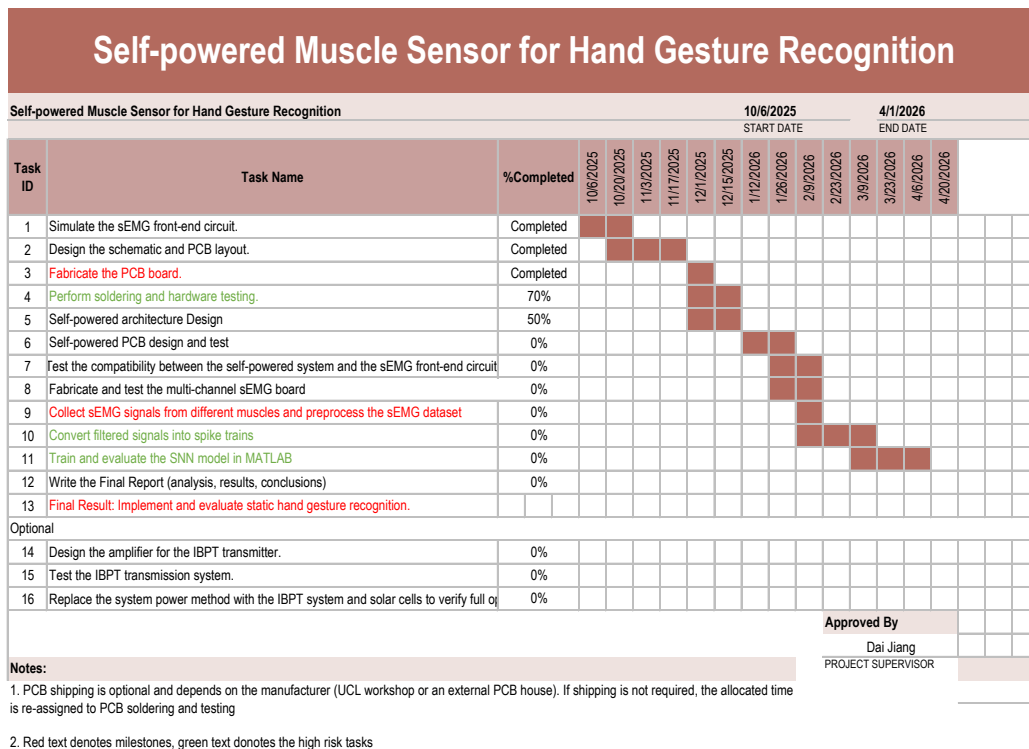


Figure 17: Gantt chart

Section 5 Future work

The remaining work will proceed along three closely related directions.

Multi-Channel sEMG analogue front-end validation The sEMG analogue front end will be fully validated on the bench and in on-body experiment. This includes resolving the INA333 amplification, re-verifying gain, bandwidth and mains-rejection performance, critically assessing whether the implemented 50 Hz notch filter provides sufficient suppression without distorting the sEMG spectrum, and then performing systematic tests on human subjects under approved ethical procedures. Building on the current single-channel prototype, the design will be extended to a three-channel front end, with both the power-input and signal-input routing optimised to reduce PCB area, and electrodes placed at appropriate spacing along the forearm so that differences in sEMG amplitude across muscle groups can be captured and mapped. Multi-channel characterisation will focus on the relationship between electrode placement and recorded muscle activity, in order to understand how small shifts in electrode position. The sEMG signals are recorded for the further data analysis.

MATLAB-based feature extraction and five-gesture recognition The future work is to build a basic hand-gesture recognition pipeline in MATLAB using the recorded multi-channel sEMG signals. The signals from each channel will be segmented into time windows, and simple time- and frequency-domain features (such as RMS value, waveform length and band-power) will be extracted. A straightforward classifier, for example LDA or SVM, will then be trained to distinguish at least five predefined hand gestures (e.g. rest, fist, pinch, wrist flexion and wrist extension). The performance of this pipeline will be evaluated using classification accuracy and confusion matrices, and its sensitivity to electrode position changes and different contraction levels will be examined.

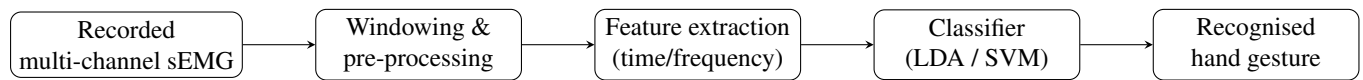


Figure 18: Planned MATLAB pipeline for hand-gesture recognition using multi-channel sEMG.

Implementation and evaluation of the self-powered architecture The self-powered architecture will be implemented and evaluated to ensure that it can provide a stable output 3.3V under self-sustained operation, meet the power requirements of the sEMG front end, and remain adaptable across different environmental conditions. As an optional extension, the existing C-IBPT prototype will be further refined and interfaced with the same power-management chain to assess its effectiveness as a supplementary power-delivery path for improving wearability and relaxing the placement constraints of the solar cells.

References

- [1] W. Li, P. Shi, and H. Yu, "Gesture recognition using surface electromyography and deep learning for prostheses hand: State-of-the-art, challenges, and future," *Frontiers in Neuroscience*, vol. Volume 15 - 2021, 2021.
- [2] A. Sultana, F. Ahmed, and M. S. Alam, "A systematic review on surface electromyography-based classification system for identifying hand and finger movements," *Healthcare Analytics*, vol. 3, p. 100126, 2023.
- [3] M. Pan, Y. Tang, and H. Li, "State-of-the-art in data gloves: A review of hardware, algorithms, and applications," *IEEE Transactions on Instrumentation and Measurement*, vol. PP, p. 1–1, 2023.
- [4] E. Fertl, E. Castillo, G. Stettinger, M. P. Cuéllar, and D. P. Morales, "Hand gesture recognition on edge devices: Sensor technologies, algorithms, and processing hardware," *Sensors*, vol. 25, no. 6, p. 1687, 2025.
- [5] M. Oudah, A. Al-Naji, and J. Chahl, "Hand gesture recognition based on computer vision: A review of techniques," *J Imaging*, vol. 6, no. 8, 2020. 2313-433x Oudah, Munir Al-Naji, Ali Chahl, Javaan Orcid: 0000-0001-6496-0543 Journal Article Review Switzerland 2021/08/31 J Imaging. 2020 Jul 23;6(8):73. doi: 10.3390/jimaging6080073.
- [6] A. Filipowska, W. Filipowski, P. Raif, M. Pieniążek, J. Bodak, P. Ferst, K. Pilarski, S. Sieciński, R. J. Doniec, J. Mieszczanin, E. Skwarek, K. Bryzik, M. Henkel, and M. Grzegorzek, "Machine learning-based gesture recognition glove: Design and implementation," *Sensors*, vol. 24, no. 18, p. 6157, 2024.
- [7] L. Pelaez Murciego, M. C. Henrich, E. G. Spaich, and S. Dosen, "Reducing the number of emg electrodes during online hand gesture classification with changing wrist positions," *Journal of NeuroEngineering and Rehabilitation*, vol. 19, no. 1, p. 78, 2022.
- [8] L. Pelaez Murciego, M. C. Henrich, E. G. Spaich, and S. Dosen, "Reducing the number of emg electrodes during online hand gesture classification with changing wrist positions," *Journal of NeuroEngineering and Rehabilitation*, vol. 19, no. 1, p. 78, 2022.
- [9] C. Lin, Y. Wang, and M. Dai, "Robust gesture recognition based on attention-deep fast convolutional neural network and surface electromyographic signals," *Frontiers in Neuroscience*, vol. Volume 18 - 2024, 2024.
- [10] P. Kaifosh, T. R. Reardon, B. D. Allen, C. Anderson, S. Arnoud, R. Arora, M. Atray, L. Awad, F. Ayerbe, C. Baker, N. Baker, A. Barachant, P. Bard, W. P. Beltran, A. Berenzweig, R. Bhasin, J. Bienkowski, S. Bittner, L. Boegner, A. Bolarinwa, D. Bosley, M. Bracaglia, M. Bräcklein, M. Brandwein, J. Bravate, M. Butler, A. J. Calhoun, C.-J. Chang, D. Chenet, J. Chester, R. Chiarito, R. Chitnis, J. Choi, W. Chun, J. Chung, J. Connors, J. Costa, M. Cramer, R. Cunningham, W. F. Cusack, N. Danielson, T. J. Davidson, B. De Araujo, B. DiMaiolo, S. Draves, A. Du, Z. Edelson, P. Enemu, M. Fahmi, N. Farsad, A. Farshchian, R. Feliz, J. Fine, E. Formento, D. Freeman, J. Fu, J.-C. Gagnon-Audet, R. Gajurel, J. Gamutan, S. Gao, J. Garcia, N. T. H. Gayraud, M. Ghani, S. Ghosh, V. Gidwani, D. Giebisch, G. Gimler, A. Gramfort, L. Grosberg, B. Gunther, N. Guo, C. Gupta, S. G. Kaya, A. Ha, K. Hadjer, C. X. Hernández, S. Hertz, C. Hewitt, D. N. Hill, K. Hong, L. Hong, H. Hou, S. Hruda, A. Hsieh, V. Hsiung, R. Huang, Y. Hui, H. Hulet, S. Islam, V. Jayaram, C. Jiang, X. Jiang, B. Juarez, J. J. Jun,

- N. Y. Jun, N. Kadakia, N. Kakar, A. Kamdar, T.-C. Kao, S. Kober, *et al.*, “A generic non-invasive neuromotor interface for human-computer interaction,” *Nature*, vol. 645, no. 8081, p. 702–711, 2025.
- [11] M.-K. Kim, M.-S. Kim, H.-B. Kwon, S.-E. Jo, and Y.-J. Kim, “Wearable triboelectric nanogenerator using a plasma-etched pdms–cnt composite for a physical activity sensor,” *RSC advances*, vol. 7, no. 76, p. 48368–48373, 2017.
- [12] Y. Liu, H. Khanbarez, M. A. Halim, A. Feeney, X. Zhang, H. Heidari, and R. Ghannam, “Piezoelectric energy harvesting for self-powered wearable upper limb applications,” *Nano Select*, vol. 2, no. 8, p. 1459–1479, 2021.
- [13] A. Kong, D. Kim, and C. Harrison, “Power-over-skin: Full-body wearables powered by intra-body rf energy,” 2024.
- [14] C. J. De Luca, “Surface electromyography: Detection and recording,” *DelSys Incorporated*, vol. 10, no. 2, p. 1–10, 2002.
- [15] E. M. Spinelli, R. Pallas-Areny, and M. A. Mayosky, “Ac-coupled front-end for biopotential measurements,” *IEEE Transactions on Biomedical Engineering*, vol. 50, no. 3, p. 391–395, 2003.
- [16] T. Instruments, “Ina333 micro-power zero-drift rail-to-rail output instrumentation amplifier datasheet,” report, Texas Instruments Incorporated, 2015.
- [17] B. Carter, “More filter design on a budget,” Tech. Rep. SLOA096, Texas Instruments, Dallas, TX, 2001.
- [18] Texas Instruments, *INA333 Micro-Power, Zero-Drift, Rail-to-Rail Out Instrumentation Amplifier*, 2024. Datasheet, Rev. C.
- [19] Analog Devices, *AD8541/AD8542/AD8544: Rail-to-Rail Input/Output, 45 μ A, Single-Supply Amplifiers*, 2023. Datasheet, Rev. H.
- [20] Texas Instruments, *bq25570 Nano Power Boost Charger and Buck Converter for Energy Harvester Powered Applications*, 2019. Rev. G, bq25570 energy-harvesting PMIC datasheet.
- [21] Microchip Technology Inc., *MCP73871 Stand-Alone System Load Sharing and Li-Ion/Li-Polymer Battery Charge Management Controller*, 2022. Rev. F, MCP73871 battery charge management controller datasheet.

Appendix

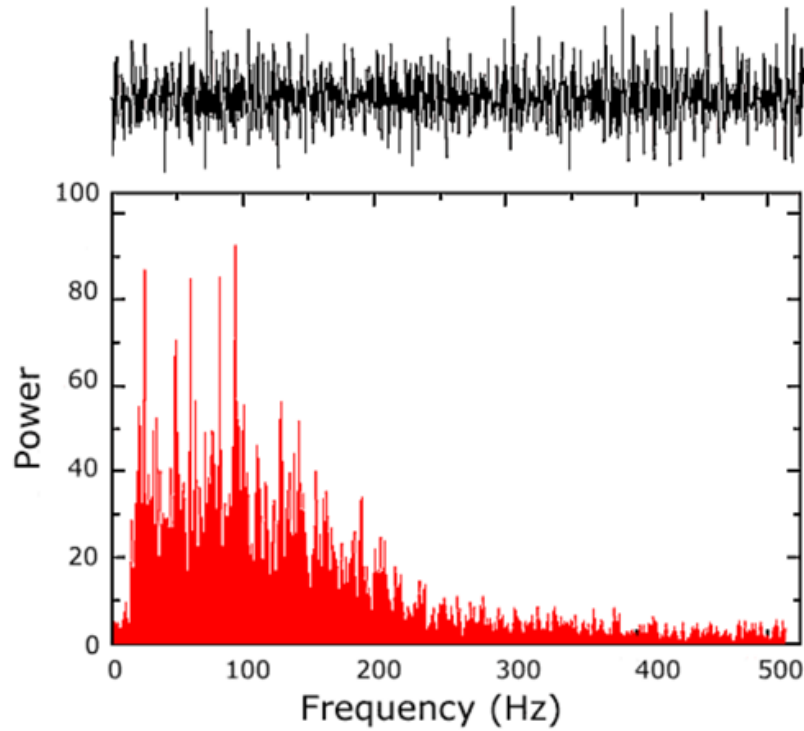
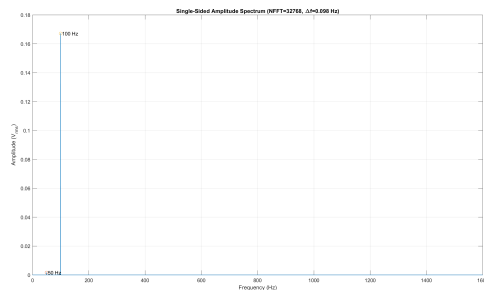
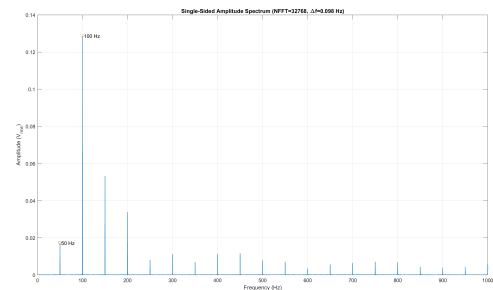


Figure A1: Frequency spectrum of the EMG signal recorded from the tibialis anterior muscle during a steady isometric contraction at 50% of maximum voluntary effort.

5.1 Additional simulation and measurement results

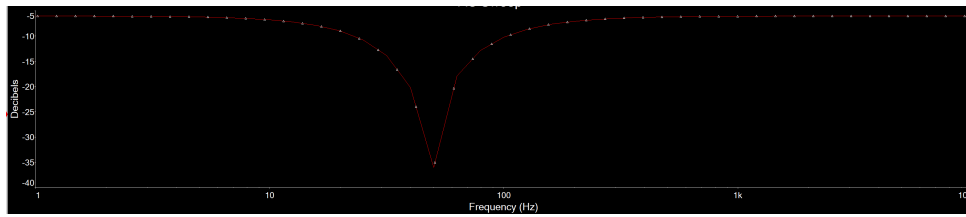


(a) With DRL: frequency spectrum.

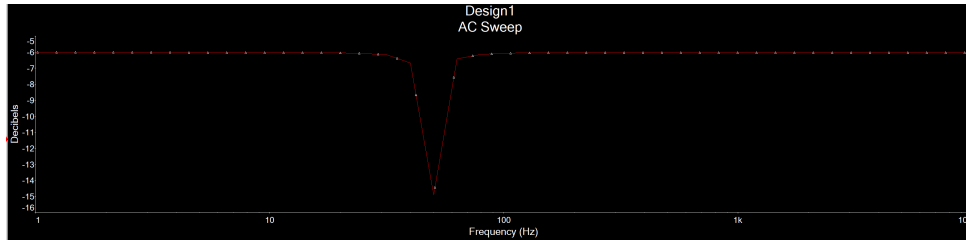


(b) Without DRL: frequency spectrum.

Figure A2: Frequency spectra with and without the DRL loop (100 Hz differential input; 50 Hz common-mode injected).



(a) Low- Q response.



(b) High- Q response.

Figure A3: Frequency-response comparison between low- Q and high- Q notch filters.

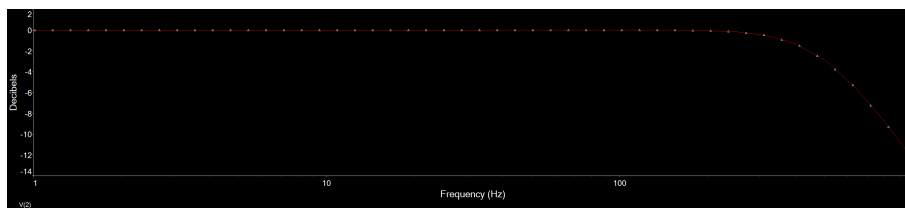


Figure A4: Frequency response of the second-order low-pass filter ($f_c \approx 494$ Hz).

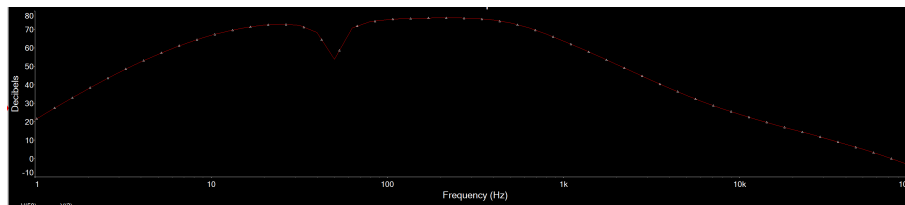


Figure A5: Overall Design Frequency Spectrum